Appendix 3A: The Spatial Population over Regional Components (SPoRC) Model: Technical Description of the Assessment Model for Sablefish

The 2026 SAFE for Alaskan sablefish uses the SPoRC package, which supports single-region, fleets-as-areas (FAA), and spatial models. The sections below outline the generalized model structure used across these assessment configurations, as well as any specific mathematical configurations pertinent to the models presented throughout. Additional details on model descriptions can also be found at https://chengmatt.github.io/SPoRC/articles/c_model_equations.html.

Population dynamics follow an annual time step, which proceed in the following sequence:

1. Recruitment and tag releases initially occur (tag releases only occur if tagging data are used),
2. Markovian movement of individuals then follow (movement only occurs in the spatial model), and
3. Total mortality, chronic tag loss (if applicable), and ageing processes occur.

These processes are modeled across four main partitions: region (r), year (y), age (a and a_+, where a_+ is the plus group), and sex (s). In single-region models, most equations reduce by setting $r = 1$.

Process Equations

Population Initialization

Initialization of equilibrium abundance differs between the single-region and spatial models due to the need to incorporate movement dynamics in the latter. In the single-region model, the initial equilibrium population is computed as:

$$N'_{r=1,a,s} = \mu^{\text{Rec}} \exp\left(-(a-1)\cdot\left(Z'_{r=1,a,s}\right)\right)\psi_s, \quad \text{for } 2 \le a < a_+$$

$$Z'_{r=1,a,s} = \text{Natmort}_{r=1,y=1,a,s} + \mu^{\text{Fsh}}_{r=1,f=1}\text{Fmort}^{\text{InitProp}}\text{Sel}^{\text{Fsh}}_{r=1,y=1,a,s,f=1}$$

where $N'_{r=1,a,s}$ are the equilibrium numbers-at-age, μ^{Rec} is a global recruitment parameter, $Z'_{r=1,a,s}$ is the initial instantaneous total mortality rate, $\text{Natmort}_{r=1,y=1,a,s}$ is the instantaneous natural mortality rate, $\mu^{\text{Fsh}}_{r=1,f=1}$ is the mean fishing mortality rate from the longline fleet (f denotes fishery fleet), $\text{Fmort}^{\text{InitProp}}$ is the proportion of the mean fishing mortality from the longline fleet applied during the initialization stage, $\text{Sel}^{\text{Fsh}}_{r=1,y=1,a,s,f=1}$ is the fishery selectivity-at-age for the longline fleet, ψ_s is the sex ratio of the equilibrium population (fixed at 50:50).

The plus group (a_+) of the initial population is then derived by solving for a geometric series:

$$N'_{r,a,s} = \mu^{\text{Rec}} \frac{\exp\left(-(a_+ - 1)\cdot\left(Z'_{r=1,a=a_+,s}\right)\right)}{1-\exp\left(Z'_{r=1,a=a_+,s}\right)}\psi_s$$

By contrast, the initial population abundance for the spatial model using the solution to a matrix geometric series to appropriately account for movement dynamics in equilibrium:

$$\boldsymbol{N'}_{a_{+},s} = \left[\mathbf{I} - \mathbf{O}_{a_{+}}\right]^{-1} \mathbf{O}_{a_{+}-1} \boldsymbol{N'}_{a_{+-1},s}$$

where $\mathbf{O}_a$ is a matrix representing the proportion of individuals surviving and moving among regions. $\mathbf{O}_a$ is defined as the matrix product of a diagonal survival matrix $\exp[\mathrm{diag}(-\mathbf{Z}_a)]$ and a movement rate matrix $\mathbf{M}_{y=1,a,s}$.

Following the definition of equilibrium age structure for both the single-region and spatial model, initial age deviations are applied:

$$N_{r,y=1,a\notin 2,s} = N'_{r,a\notin 2,s} \exp\left(\epsilon_{r,i}^{\mathrm{Init}}\right)$$

$N_{r,y=1,a\notin 2,s}$ are the stochastic numbers-at-age in the first year, and initial age deviations ($\epsilon_{r,i}$) are constrained by a lognormal distribution.

Recruitment Processes

In both model types, recruitment is estimated based on lognormal deviations about a mean recruitment parameter:

$$N_{r,y,a=2,s} = \mu^{\mathrm{Rec}} \exp\left(\epsilon_{r,y}^{\mathrm{Rec}} - \frac{\sigma_{\mathrm{Rec}}^2}{2} b_y\right) \psi_s \zeta_r$$

Here, $\epsilon_{r,y}$ are annual, lognormally distributed recruitment deviations with a lognormal bias correction term ($\frac{\sigma_{\mathrm{Rec}}^2}{2} b_y$).

Population Projection

Following recruitment processes, the population can then be projected forward. In the context of the spatial model, Markovian movement dynamics are first applied:

$$\mathbf{N}_{y,a,s} = \left(\mathbf{N}_{y,a,s}\right)^{\mathrm{T}} \mathbf{M}_{y,a,s}$$

After the population undergoes movement, individuals experience mortality and ageing processes via an exponential mortality model:

$$N_{r,y+1,a+1,s} = N_{r,y,a,s} \exp\left(-Z_{r,y,a,s}\right), \quad \text{for } 2 \leq a < a_{+}$$
$$N_{r,y+1,a,s} = N_{r,y,a-1,s} \exp\left(-Z_{r,y,a-1,s}\right) + N_{r,y,a,s} \exp\left(-Z_{r,y,a,s}\right), \quad \text{for } a = a_{+}$$

$Z_{r,y,a,s}$ denotes the annual total instantaneous mortality rate and is defined as:

$$Z_{r,y,a,s} = \text{Natmort}_{r,y,a,s} + \sum_{f} \text{Sel}^{\text{Fsh}}_{r,y,a,s,f} \, \text{Fmort}_{r,y,f}$$

$\text{Fmort}_{r,y,f}$ is the annual instantaneous fishing mortality rate parameterized based on lognormal deviations about a mean fishing mortality parameter for a given fishery fleet:

$$\text{Fmort}_{r,y,f} = \mu^{\text{Fsh}}_{r,f} \exp\left(\epsilon^{\text{Fsh}}_{r,y,f}\right).$$

Observation Equations

Fishery Observation Model

The fishery observation model describes the expected catch-at-age, catch-at-length, catch (in units of biomass), and fishery indices. In cases where the fishery is defined as a FAA model, the f index simply extends and the following equations can be generalized. Expected catch-at-age ($C^{a}_{r,y,a,s,f}$) is computed using Baranov's catch equation:

$$C^{a}_{r,y,a,s,f} = \frac{\text{Fmort}_{r,y,f} \text{Sel}^{\text{Fsh}}_{r,y,a,s,f}}{Z_{r,y,a,s}} N_{r,y,a,s} \left[1 - \exp\left(-Z_{r,y,a,s}\right)\right]$$

Catch-at-length ($C^{l}_{r,y,l,s,f}$) can then be derived using the following equation:

$$C^{l}_{r,y,l,s,f} = \left(\mathbf{A}^{l}_{r,y,s}\right)^{\mathrm{T}} \mathbf{C^a}_{r,y,s,f}$$

where $\mathbf{A}^{l}_{r,y,s}$ is the size-age transition matrix. Expected catch ($\text{Catch}_{r,y,f}$) is then computed summation of the expected catch-at-age across ages and sexes, multiplied by their respective weights-at-age ($W_{r,y,a,s}$):

$$\text{Catch}_{r,y,f} = \sum_{a}^{a_+} \sum_{s}^{n_s} C^{a}_{r,y,a,s,f} \, W_{r,y,a,s}$$

The expected fishery index ($\text{FshIdx}_{r,y,f}$) is computed as:

$$\text{FshIdx}_{r,y,f} = q^{\text{Fsh}}_{r,y,f} \sum_{a}^{a_+} \sum_{s}^{n_s} N_{r,y,a,s} \exp\left(-Z_{r,y,a,s} 0.5\right) \text{Sel}^{\text{Fsh}}_{r,y,a,s,f} W_{r,y,a,s}$$

where $q^{\text{Fsh}}_{r,y,f}$ is the catchability coefficient for a given fishery fleet.

Survey Observation Model

Similarly, the survey observation model describes the expected survey catch-at-age, survey catch-at-length, and survey indices. Moreover, if surveys are defined as an FAA model, the *sf* index similarly extends, and the following equations can also be generalized. Expected survey catch-at-age ($I^a_{r,y,a,s,sf}$) is calculated as follows:

$$I^a_{r,y,a,s,sf} = N_{r,y,a,s}\exp\left(-Z_{r,y,a,s}0.5\right)\mathrm{Sel}^{\mathrm{Srv}}_{r,y,a,s,sf}$$

where subscript sf denotes a given survey fleet and $\mathrm{Sel}^{\mathrm{Srv}}_{r,y,a,s,sf}$ is the survey selectivity-at-age pattern. Expected survey catch-at-length ($I^l_{r,y,l,s,sf}$) is given by:

$$I^l_{r,y,l,s,sf} = \left(\mathbf{A}^l_{r,y,s}\right)^{\mathrm{T}}\mathbf{I^a}_{r,y,s,f}$$

Survey indices ($\mathrm{SrvIdx}_{r,y,sf}$) is computed as either abundance-based or biomass-based. Abundance-based survey indices are calculated as:

$$\mathrm{SrvIdx}_{r,y,sf} = q^{\mathrm{Srv}}_{r,y,sf}\sum_{a}^{a_+}\sum_{s}^{n_s} I^a_{r,y,a,s,sf}$$

while biomass-based indices are computed as:

$$\mathrm{SrvIdx}_{r,y,sf} = q^{\mathrm{Srv}}_{r,y,sf}\sum_{a}^{a_+}\sum_{s}^{n_s} I^a_{r,y,a,s,sf}\, W_{r,y,a,s}$$

In both equations, $q^{\mathrm{Srv}}_{r,y,sf}$ represents the survey catchability coefficient.

Selectivity Models

Several parameterizations of both fishery and survey selectivity are used throughout. In general, most fleets assume logistic selectivity, which is parameterized as:

$$\mathrm{Sel}_a = \frac{1}{1+\exp\left(-k(a-a_{50\%})\right)}$$

where k is the slope of the logistic function and $a_{50\%}$ denotes the age in which 50% of individuals are selected.

The other common selectivity form specified for certain model formulations presented includes a dome-shaped gamma selectivity form:

$$p = 0.5\left[\sqrt{a^{max}+4\gamma^2} - a^{max}\right]$$

$$\mathrm{Sel}_a = \left(\frac{a}{a^{max}}\right)^{\frac{a^{max}}{p}}\exp\left(\frac{a^{max}-b}{p}\right)$$

In this parameterization, p is a derived power parameter, γ is the shape parameter that describes the steepness of the descending limb, and a^{max} describes the age-at-maximum selection.

As part of sensitivity analyses presented for both single-region and FAA models, certain models also specified continuous semi-parametric time-varying selectivity in attempts to address time-varying changes in availability. These models assumed logistic selectivity as the parametric form, with independent and identically distributed (iid) semi-parametric lognormal deviations by age $\left(\epsilon_{y,a}^{Sel}\right)$from the base parametric form and were configured as:

$$\mathrm{Sel}'_{y,a} = \frac{1}{1+\exp\left(-k(a-a_{50\%})\right)}\exp\left(\epsilon_{y,a}^{\mathrm{Sel}}\right)$$

$$\mathrm{Sel}_{y,a} = \frac{\mathrm{Sel}'_{y,a}}{\mathrm{mean}(\mathbf{Sel}')}$$

where deviations were applied about the underlying parametric logistic form and selectivity values were mean-standardized to improve interpretability and maintain identifiability among years. Sex- and fleet-specific subscripts were also estimated for selectivity deviations, but are omitted from the notation herein for brevity. Additional smoothing penalties were imposed to maintain identifiability and prevent large year-to-year and age-to-age changes in selectivity values, thereby encouraging gradual temporal and age transitions in selectivity patterns (additional details are provided in the Likelihoods section).

Tagging Observation Model

The tagging observation model tracks tag cohorts ($T_{r,y,a,s}^{k}$) by the combination of release region and release year (k) and follows a Brownie tag attrition framework. Tag cohorts are tracked for a pre-defined maximum duration (maximum tag liberty; n_L), after which calculations for the tag cohort are no longer computed. This is done to help with computation time. Additionally, tag cohorts also have the option to undergo a fraction of mortality in the year of release, such that if a given cohort is released during the middle of the year, mortality processes are discounted. If mortality discounting is specified, movement does not occur in the year of release. In general, the process dynamics for the tagged cohort mimics those specified for the overall population.

Immediately following release, tag cohorts are decremented by an initial tag-induced mortality rate:

$$T_{r,y,a,s}^{k} = T_{r,y,a,s}^{k}\exp(-\tau)$$

where τ is the initial tag-induced mortality rate. If tagged cohorts are released at the beginning of a calendar year, Markovian movement occurs (movement does not occur otherwise):

$$\mathbf{T}_{y,a,s}^{k} = \left(\mathbf{T}_{y,a,s}^{k}\right)^{\mathrm{T}}\mathbf{M}_{y,a,s}$$

Mortality and ageing of the tagged cohort then occur:

$$T^k_{r,y+1,a+1,s} = T^k_{r,y,a,s}\exp\left(-Z^{\mathrm{Tag}}_{r,y,a,s}\right), \quad \text{for } 2 \le a < a_+$$
$$T^k_{r,y+1,a,s} = T^k_{r,y,a-1,s}\exp\left(-Z^{\mathrm{Tag}}_{r,y,a-1,s}\right) + T^k_{r,y,a,s}\exp\left(-Z^{\mathrm{Tag}}_{r,y,a,s}\right), \quad \text{for } a = a_+$$

where tagged cohorts follow an exponential mortality model, with accumulation of individuals in the plus-group. Total mortality for the tagged cohort ($Z^{\mathrm{Tag}}_{r,y,a,s}$) is specified as:

$$Z^{\mathrm{Tag}}_{r,y,a,s} = \left(\kappa + \mathrm{NatMort}_{r,y,a,s} + \sum_f \mathrm{Sel}^{\mathrm{Fsh}}_{r,y,a,s,f}\ \mathrm{Fmort}_{r,y,f}\right)$$

where κ is a parameter describing chronic tag loss (i.e., annual tag shedding). Similar to computations for catch-at-age, tag recaptures are calculated using a modified version of Baranov's catch equation:

$$\mathrm{Recap}^k_{r,y,a,s} = \beta_{r,y}\frac{\sum_f \mathrm{Sel}^{\mathrm{Fsh}}_{r,y,a,s,f}}{Z^{\mathrm{Tag}}_{r,y,a,s}} T^k_{r,y,a,s}\left[1 - \exp\left(-Z^{\mathrm{Tag}}_{r,y,a,s}\right)\right]$$

$\beta_{r,y}$ represents a tag reporting rate parameter that can vary across space and time, and is estimated in logit space such that it is constrained between $[0,1]$.

Likelihoods

The model incorporates likelihood components for the following data sources:

1. fishery catches,
2. fishery indices,
3. fishery age compositions,
4. fishery length compositions,
5. survey indices,
6. survey age compositions,
7. survey length compositions, and
8. tagging data.

The total likelihood is the sum of the individual likelihood contributions from these sources, along with penalties and priors. This total is minimized using a non-linear optimization algorithm to estimate model parameters.

Observation Likelihoods

Fishery catches are fit using a lognormal likelihood. The log-likelihood for observed catch, $L\left(\log\left(\mathrm{ObsCatch}_{r,y,f}\right)\right)$, is defined as:

$$L\left(\log\left(\text{ObsCatch}_{r,y,f}\right)\right) = \lambda_{\text{ObsCatch}_{r,y,f}} \cdot \frac{1}{\sqrt{2\pi}\,\sigma_{\text{ObsCatch}_{r,y,f}}} \exp\left(-\frac{\left[\log\left(\text{ObsCatch}_{r,y,f}\right) - \log\left(\text{Catch}_{r,y,f}\right)\right]^2}{2\sigma^2_{\text{ObsCatch}_{r,y,f}}}\right)$$

Here, $\lambda_{\text{ObsCatch}_{r,y,f}}$ is the likelihood weight, $\text{ObsCatch}_{r,y,f}$ is the observed catch, $\text{Catch}_{r,y,f}$ is the predicted catch, and $\sigma^2_{\text{ObsCatch}_{r,y,f}}$ is the variance on the log scale.

Fishery indices are also fit using a lognormal likelihood. The log-likelihood for observed indices is:

$$L\left(\log\left(\text{ObsFshIdx}_{r,y,f}\right)\right) = \lambda_{\text{ObsFshIdx}_{r,y,f}} \cdot \frac{1}{\sqrt{2\pi}\,\sigma_{\text{ObsFshIdx}_{r,y,f}}} \exp\left(-\frac{\left[\log\left(\text{ObsFshIdx}_{r,y,f}\right) - \log\left(\text{FshIdx}_{r,y,f}\right)\right]^2}{2\sigma^2_{\text{ObsFshIdx}_{r,y,f}}}\right)$$

The parameter $\lambda_{\text{ObsFshIdx}_{r,y,f}}$ controls the weight of fishery indices to the objective function. $\text{ObsFshIdx}_{r,y,f}$ represents the observed fishery indices, and $\sigma^2_{\text{ObsFshIdx}_{r,y,f}}$ denotes the variance of the index.

Similarly, survey indices are fit to assuming a lognormal likelihood, which is given by:

$$L\left(\log\left(\text{ObsSrvIdx}_{r,y,sf}\right)\right) = \lambda_{\text{ObsSrvIdx}_{r,y,sf}} \cdot \frac{1}{\sqrt{2\pi}\,\sigma_{\text{ObsSrvIdx}_{r,y,sf}}} \exp\left(-\frac{\left[\log\left(\text{ObsSrvIdx}_{r,y,sf}\right) - \log\left(\text{SrvIdx}_{r,y,sf}\right)\right]^2}{2\sigma^2_{\text{ObsSrvIdx}_{r,y,sf}}}\right)$$

Again, $\lambda_{\text{ObsSrvIdx}_{r,y,sf}}$ is the likelihood weight applied to survey indices, $\text{ObsSrvIdx}_{r,y,sf}$ are the observed survey indices, and $\sigma^2_{\text{ObsSrvIdx}_{r,y,sf}}$ indicates the variance of the index.

Several formulations of composition likelihoods were explored throughout. Across most models, both fishery and survey compositions were generally fitted assuming a multinomial likelihood, which is given by:

$$L(\left(\text{ObsCompositionData}_{r,y,j}\right) = \lambda_{\text{ObsCompositionData}_{r,y,j}} \cdot \text{ISS}_{r,y,j} \cdot \prod_{b=1}^{n_b} E_{r,y,b,j}^{O_{r,y,b,j}}$$

where subscript j is used to indicate a fishery or survey fleet. $\lambda_{\text{ObsCompositionData}_{r,y,j}}$ are likelihood weights applied to composition data (can be derived using iterative weighting methods), $ISS_{r,y,j}$ is the input sample size, the b subscript generically indicates a bin number, $E_{r,y,b,j}$ denotes the expected composition proportions, and $O_{r,y,b,j}$ are the observed composition proportions.

In some cases, the Dirichlet-multinomial was also employed to better understand the impact of self-weighting likelihoods on model results, and was parameterized as:

$$L\left(\text{ObsCompositionData}_{r,y,j}\right) =$$
$$\lambda_{\text{ObsCompositionData}_{r,y,j}} \cdot \frac{\Gamma\left(\text{ISS}_{r,y,j}+1\right)}{\text{ISS}_{r,y,j} \cdot O_{r,y,b,j}+1} \cdot \frac{\Gamma\left(\text{ISS}_{r,y,j}\theta_{r,j}\right)}{\Gamma(\text{ISS}_{r,y,j}+\text{ISS}_{r,y,j}\theta_{r,j})} \cdot$$
$$\prod_{b=1}^{n_b} \frac{\Gamma(\text{ISS}_{r,y,j}O_{r,y,b,j}+\text{ISS}_{r,y,j}\theta_{r,j}E_{r,y,b,j})}{\text{ISS}_{r,y,j}\theta_{r,j}E_{r,y,b,j}}$$

Here, $\theta_{r,j}$ is the overdispersion parameter of the Dirichlet-multinomial that adjusts the input sample size. The effective sample size ($\text{ESS}_{r,y,j}$) can then be derived as:

$$\text{ESS}_{r,y,j} = \frac{1}{1+\theta_{r,j}} + \text{ISS}_{r,y,j}\frac{\theta_{r,j}}{1+\theta_{r,j}}$$

Therefore, if $\theta_{r,j}$ was large such that $\theta_{r,j} \gg \text{ISS}_{r,y,j}$, then $\text{ESS}_{r,y,j} \rightarrow \text{ISS}_{r,y,j}$. By contrast, if $\theta_{r,j}$ was estimated at a small value such that $\theta_{r,j} \ll \text{ISS}_{r,y,j}$, then $\theta_{r,j}$ approximates the ratio of $\text{ESS}_{r,y,j}$ and $\text{ISS}_{r,y,j}$.

Within the SPoRC framework, compositions can be arranged to be split by sex or joint by sex. For brevity, we will describe the joint by sex approach, which is the author recommended approach (i.e., models 25.7 +). Here, composition proportions are assumed to sum to one jointly across both ages and sexes. Thus, for the expected fishery age compositions, this is computed as:

$$E_{r,y,b,j} = \frac{\left[C^a_{r,y,a,s=1,f},\ C^a_{r,y,a,s=2,f}\right]}{\sum_a \sum_s C^a_{r,y,a,s,f}}$$

with $b \in \{1, \dots, n_a \cdot n_s\}$. The expected survey age compositions are computed similarly as:

$$E_{r,y,b,j} = \frac{\left[I^a_{r,y,a,s=1,sf},\ I^a_{r,y,a,s=2,sf}\right]}{\sum_a \sum_s I^a_{r,y,a,s,sf}}$$

For expected fishery length compositions, this is given by:

$$E_{r,y,b,j} = \frac{\left[C^l_{r,y,l,s=1,f},\ C^l_{r,y,l,s=2,f}\right]}{\sum_l \sum_s C^a_{r,y,l,s,f}}$$

where $b \in \{1, \dots, n_l \cdot n_s\}$. Expected survey length compositions follow in a similar fashion:

$$E_{r,y,b,j} = \frac{\left[I^{l}_{r,y,l,s=1,f},\ I^{l}_{r,y,l,s=2,f}\right]}{\sum_l \sum_s I^{a}_{r,y,l,s,f}}$$

To fit tagging data, a multinomial release-conditioned likelihood is assumed, which describes both recaptured and non-recaptured states. The observed and expected recapture proportions are defined as:

$$\begin{aligned}
\text{PropRecap}^{k}_{r,y,a,s} &= \frac{\text{Recap}^{k}_{r,y,a,s}}{\text{InitTag}^{k}} \\
\text{ObsPropRecap}^{k}_{r,y,a,s} &= \frac{\text{ObsRecap}^{k}_{r,y,a,s}}{\text{InitTag}^{k}}
\end{aligned}$$

where InitTag^{k} is the total number of tags released for tag cohort k.

The corresponding non-recapture proportions are computed as:

$$\begin{aligned}
\text{PropNonRecap}^{k} &= 1 - \sum_{r}\sum_{y}\sum_{a}\sum_{s} \text{PropRecap}^{k}_{r,y,a,s} \\
\text{ObsPropNonRecap}^{k} &= 1 - \sum_{r}\sum_{y}\sum_{a}\sum_{s} \text{ObsPropRecap}^{k}_{r,y,a,s}
\end{aligned}$$

Expected and observed tagging outcomes are represented by the following vectors:

$$\begin{aligned}
\mathbf{O}^{k}_{\text{Tagging}} &= \left\{\text{ObsPropRecap}^{k}_{r,y,a,s},\ \text{ObsPropNonRecap}^{k}\right\} \\
\mathbf{E}^{k}_{\text{Tagging}} &= \left\{\text{PropRecap}^{k}_{r,y,a,s},\ \text{PropNonRecap}^{k}\right\}
\end{aligned}$$

A multinomial likelihood is then used to evaluate tagging data:

$$L^{k}_{\text{Tagging}} = \lambda_{\text{Tagging}} \cdot \text{InitTag}^{k} \prod_{i} \left(E^{k}_{i,\text{Tagging}}\right)^{O^{k}_{i,\text{Tagging}}}$$

Here, λ_{Tagging} is the likelihood weight, and i indexes elements of the observed and expected tagging proportion vectors.

Several priors and penalties are applied to constrain, stabilize, and inform parameter estimation. Specifically, priors are used for natural mortality, selectivity, movement, recruitment proportions, and tag reporting rates. The last three parameter types are only applicable for the spatial model. Additionally, penalties are used to constrain the estimation of parameters associated with deviations, which include: initial ages, recruitment, and fishing mortality.

In the case of natural mortality, a lognormal prior is utilized:

$$P\left(\log(\mathrm{NatMort}_s)\right) = \frac{1}{\sqrt{2\pi}\,\sigma_{\mathrm{NatMort}}} \exp\left(-\frac{[\log(\mathrm{NatMort}_s) - \log(\mu_s^{\mathrm{NatMort}})]^2}{2\sigma_{\mathrm{NatMort}}^2}\right)$$

The variance of the prior is given by $\sigma_{\mathrm{NatMort}}^2$, and μ_s^{NatMort} denotes the prior mean.

Selectivity priors are also utilized which serve as regularizing priors to facilitate stable parameter estimation. These priors are assumed to be lognormal and are applied to the selectivity parameters themselves:

$$P\left(\chi_{r,p,t,s,j}\right) = \frac{1}{\chi_{r,p,t,s,j}\sqrt{2\pi}\,\sigma_{r,p,t,s,j}^{\mathrm{Sel}}} \exp\left(-\frac{\left[\ln\left(\chi_{r,p,t,s,j}\right) - \ln\left(\mu_{r,p,t,s,j}^{\mathrm{Sel}}\right)\right]^2}{2\left(\sigma_{r,p,t,s,j}^{\mathrm{Sel}}\right)^2}\right)$$

where $\chi_{r,p,t,s,j}$ is the selectivity parameter for region r, parameter p, time period t, sex s, and fleet j (fishery or survey). $\mu_{r,p,t,s,j}^{\mathrm{Sel}}$ is the prior mean, while $\left(\sigma_{r,p,t,s,j}^{\mathrm{Sel}}\right)^2$ is the prior variance.

Priors on movement values are assumed to arise from a Dirichlet process:

$$P\left(\mathbf{M}_{k,y,a,s}\right) = \frac{\Gamma\left(n_r \cdot \alpha_{y,a,s}\right)}{\left[\Gamma\left(\alpha_{y,a,s}\right)\right]^{n_r}} \prod_{r=1}^{n_r} M_{k,r,y,a,s}^{\alpha_{y,a,s}-1}$$

where $\mathbf{M}_{k,y,a,s} = \{M_{k,1,y,a,s}, \dots, M_{k,n_r,y,a,s}\}$ is the vector of movement probabilities from region k to all regions, and $\alpha_{y,a,s}$ is the concentration parameter controlling the strength of the prior for year y, age a, and sex s.

Regional recruitment is derived by apportioning a global recruitment parameter using regional recruitment proportions (i.e., $\mu^{\mathrm{Rec}} \cdot \zeta_r$). Here, ζ_r is derived via a multinomial logit transformation and Dirichlet priors are used to help constrain estimation:

$$P(\boldsymbol{\zeta}) = \frac{\Gamma(n_r \cdot c)}{[\Gamma(c)]^{n_r}} \prod_{r=1}^{n_r} \zeta_r^{c-1}$$

$\boldsymbol{\zeta} = \{\zeta_1, \zeta_2, \dots, \zeta_{n_r}\}$ are the estimated recruitment proportions across regions, and c is a shared concentration parameter governing the spread of the Dirichlet distribution.

To facilitate the estimation of tag reporting rates, a symmetric beta distribution is applied:

$$P\left(\beta_{r,y}\right) = \left(\beta_{r,y} + 1e - 4\right)^{\beta_{\mathrm{PriorScale}}} \left(1 - \beta_{r,y} + 1e - 4\right)^{\beta_{\mathrm{PriorScale}}}$$

$\beta_{\mathrm{PriorScale}}$ determines the scale of the tag reporting parameter and determines how strongly to penalize estimates when they approach the bounds of [0,1].

Estimation of initial age deviations $\epsilon_{r,i}^{\text{Init}}$ is constrained by the following normal likelihood:

$$L\left(\epsilon_{r,i}^{\text{Init}}\right) = \frac{1}{\sqrt{2\pi}\,\sigma_{\text{Init}}} \exp\left(-\frac{\left(\epsilon_{r,i}^{\text{Init}}\right)^2}{2\sigma_{\text{Init}}^2}\right)$$

with a mean of 0 and a variance determined by σ_{Init}^2.

Similarly, estimation of recruitment deviations is also constrained by a normal likelihood with a mean of 0 and a variance of σ_{Rec}^2. This is given by:

$$L\left(\epsilon_{r,y}^{\text{Rec}}\right) = \frac{1}{\sqrt{2\pi}\,\sigma_{\text{Rec}}} \exp\left(-\frac{\left(\epsilon_{r,y}^{\text{Rec}}\right)^2}{2\sigma_{\text{Rec}}^2}\right)$$

Estimation of fishing mortality deviations is also constrained by a normal likelihood. Fishing mortality deviations are then assumed to have a mean of 0 and a variance of σ_{Fsh}^2:

$$L\left(\epsilon_{r,y,f}^{\text{Fsh}}\right) = \frac{1}{\sqrt{2\pi}\,\sigma_{\text{Fsh}}} \exp\left(-\frac{\left(\epsilon_{r,y,f}^{\text{Fsh}}\right)^2}{2\sigma_{\text{Fsh}}^2}\right)$$

In model configurations where semi-parametric time-varying selectivity is specified, estimation of semi-parametric deviations is also constrained by a normal likelihood. Semi-parametric deviations are assumed to then have a mean of 0 and a variance of σ_{Sel}^2:

$$L\left(\epsilon_{r,y,a}^{\text{Sel}}\right) = \frac{1}{\sqrt{2\pi}\,\sigma_{\text{Sel}}} \exp\left(-\frac{\left(\epsilon_{r,y,a}^{\text{Sel}}\right)^2}{2\sigma_{\text{Sel}}^2}\right)$$

Additionally, second-order smoothing penalties are also imposed to ensure that selectivity values do not drastically vary by temporally or across ages:

$$P_{\text{SelBinCurve}_r} = \sum_{y=1}^{n_y} \sum_{b=2}^{n_b-1} \left(\log\left(\text{Sel}_{r,y,a+1}\right) - 2\log\left(\text{Sel}_{r,y,a}\right)\log\left(Sel_{r,y,a-1}\right)\right)^2$$

$$P_{\text{SelYrCurve}_r} = \sum_{y=2}^{n_y-1} \sum_{b=1}^{n_b} \left(\log\left(\text{Sel}_{r,y+1,a}\right) - 2\log\left(\text{Sel}_{r,y,a}\right)\log\left(\text{Sel}_{r,y-1,a}\right)\right)^2$$

The penalties associated with selectivity also include sex- and fleet-specific subscripts, although these are omitted from the notation herein for brevity.

Lastly, the joint likelihood to be minimized represents the sum of all observational likelihood components, priors, and penalties defined above:

$$\text{Joint Likelihood} = \sum \text{Observation Likelihoods} + \sum \text{Priors} + \sum \text{Penalties}$$

Note that some of these components may be zero depending on the configuration of the model (e.g., tagging likelihood would equal 0 in a single-region model).